

Supervisor Agent

Role: Orchestration Layer · Pipeline Position: Orchestrator · Endpoint: POST /supervisor/chat

Parses natural language commands via Groq LLM and orchestrates the full multi-agent pipeline.

Overview

The Supervisor Agent sits above all five domain agents. It provides a chat interface that accepts free-form natural language delivery instructions, uses the Groq AI model to extract structured intent, maps city names to coordinates, and routes execution to the correct workflow. It also maintains per-session conversation memory and tracks end-to-end execution latency.

API Endpoints

Method	Path	Description
POST	/supervisor/chat	Send a natural language command to trigger a workflow
POST	/supervisor/execute	Directly execute a named workflow with structured parameters
GET	/supervisor/status	Get supervisor and registered agent status

LLM Intent JSON Schema

```
{
  "intent":      "dispatch | complete_trip | maintenance_status | agent_status | general",
  "workflow":    "fleet_delivery | complete_trip | vehicle_maintenance | ...",
  "pickup":      "string (optional) – origin city or lat,lon",
  "destination": "string (optional) – destination city or lat,lon",
  "weight":      "number (optional) – cargo weight kg",
  "vehicle_number": "string (optional) – registration plate"
}
```

City → Coordinate Resolver

City Name	Resolved Coordinate
chennai	13.0827, 80.2707
coimbatore	11.0168, 76.9558
bangalore / bengaluru	12.9715, 77.5945
mumbai	19.0760, 72.8777
delhi	28.6139, 77.2090

Supported Workflows

Workflow Key	Description	Agents Triggered
fleet_delivery	Full end-to-end cargo dispatch	Dispatch → Route → Maintenance → Analytics
complete_trip	Mark trip as completed, update status	Trip completion sub-flow
vehicle_maintenance	Trigger maintenance check for a vehicle	Maintenance Agent
agent_status	Return live agent registry status	Status-only
general	General query answered by LLM directly	LLM response only

Internal Logic Flow

```
chat_workflow(message)
  ■
  ■■■ 1. LLMFactory.get('groq')           → resolve Groq adapter
  ■■■ 2. memory.add('system', prompt)      → append to conversation
  ■■■ 3. memory.add('user', message)       → append user turn
  ■■■ 4. groq_adapter.complete(messages)   → get JSON intent string
  ■■■ 5. Parse JSON intent → workflow, pickup, destination, weight
  ■■■ 6. resolve_location(pickup/destination) → city → lat,lon
  ■■■ 7. get_workflow(workflow_key)        → load from registry
  ■■■ 8. workflow.run(db, params)          → execute pipeline
  ■■■ 9. Return results + execution_time_ms
```

Module Files

File	Purpose
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agent.py	Agent registry entry
service.py	Orchestration: LLM parsing, workflow routing, response formatting
routes.py	FastAPI router — /supervisor/chat, /supervisor/execute, /supervisor/status
schemas.py	Pydantic request/response models
prompts.py	System prompt templates for the LLM
tools.py	LLM function-call tool definitions

Error Cases

Condition	Exception	HTTP Code
Groq adapter fails to load	Exception re-raised	500
LLM returns malformed JSON	AIParserException	422
Unknown workflow key	AgentFleetException	400
Sub-agent pipeline failure	Exception from sub-agent	Varies